E-LEARNING MODULE (CORE ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Teachers • IT Team • Students • Parents
Related Modules: Student Management • Academics • Curriculum • Examinations • Communication • Parent Portal

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1. MODULE OVERVIEW

1.1 Purpose

The E-Learning Module provides a centralized digital learning environment for delivering curriculum-aligned content, assessments, and interactive sessions. It enables schools to:

- Deliver structured online courses aligned with CBC and other curricula
- Provide learning materials and multimedia resources
- Conduct quizzes, assignments, and live sessions
- Track learner progress and engagement
- Enable teacher, student, and parent interaction

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2. CORE OBJECTIVES

1. Deliver curriculum-aligned digital learning content
2. Enable continuous assessment through quizzes and assignments
3. Track student progress and course completion
4. Support live and interactive learning sessions
5. Provide analytics for teachers, administrators, and parents
6. Ensure accessibility via mobile and offline support
7. Integrate seamlessly with academic and student management systems
8. Maintain secure and compliant data handling

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Course Created → Content Uploaded → Course Published → Students Enrolled → Lessons Accessed → Assessments Completed → Progress Tracked → Reports Generated

3.2 Daily Operational Flow

Teacher uploads or updates course content
↓
Student logs into portal and accesses lessons
↓
Student completes quizzes/assignments
↓
System tracks progress and performance
↓
Teacher reviews analytics and provides feedback
↓
Parent receives progress updates (if enabled)

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4. MODULE ARCHITECTURE

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E-Learning Module
│
├── Course Management
├── Content & Resource Library
├── Assessments (Quizzes & Assignments)
├── Live Classes & Interactive Sessions
├── Progress Tracking & Analytics
├── Teacher & Parent Dashboards
├── Student Learning Portal
├── Notifications & Communication
├── Mobile & Offline Access
├── Accessibility Features
├── Security & Role-Based Access
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5. COURSE MANAGEMENT

5.1 Features

- Create and manage courses by subject and grade
- Publish/unpublish courses for student access
- Organize courses into modules and lessons
- Track enrollment and participation

5.2 Data Model

Course

- course_id
- title
- subject_id
- grade_level
- teacher_id
- status (draft/published)
- enrollment_count

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6. CONTENT & RESOURCE LIBRARY

6.1 Features

- Upload PDFs, videos, slides, and interactive content
- Link resources to specific lessons or topics
- Categorize content by subject and grade
- Support multimedia learning (video, audio, simulations)

6.2 Data Model

Content

- content_id
- course_id
- title
- type (video, PDF, interactive, etc.)
- file_url
- duration
- uploaded_by

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7. ASSESSMENTS (QUIZZES & ASSIGNMENTS)

7.1 Features

- Create quizzes and assignments per lesson or module
- Automated grading and feedback
- Track student performance per assessment
- Support formative and summative evaluation

7.2 Data Model

Quiz

- quiz_id
- course_id
- questions
- max_score
- passing_score

Submission

- submission_id
- student_id
- quiz_id
- score
- feedback
- timestamp

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8. LIVE CLASSES & INTERACTIVE SESSIONS

8.1 Features

- Schedule live classes via Zoom/Google Meet integration
- Real-time Q&A and discussions
- Attendance tracking for live sessions
- Session recordings for later access

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9. PROGRESS TRACKING & ANALYTICS

9.1 Features

- Track course completion percentages
- Monitor student engagement and activity
- Identify at-risk learners
- Provide teacher dashboards with performance insights

9.2 Dashboards

- Course completion trends
- Student performance analytics
- Engagement metrics (logins, activity, submissions)

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10. TEACHER & PARENT DASHBOARDS

10.1 Features

- Teacher view of student progress and performance
- Parent access to child’s learning progress
- Notifications for incomplete lessons or low performance

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11. STUDENT LEARNING PORTAL

11.1 Features

- Personalized course dashboard
- Access to lessons, materials, and assessments
- Progress tracking and feedback view
- Notifications for deadlines and live sessions

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12. NOTIFICATIONS & COMMUNICATION

12.1 Features

- Alerts for new content, assignments, and deadlines
- Notifications for live classes and updates
- Integration with messaging and communication modules

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13. ACCESSIBILITY & INCLUSIVITY

13.1 Features

- Text-to-speech support
- Subtitles for video content
- Adaptive learning paths
- Multi-language support

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14. MOBILE-FIRST & OFFLINE DESIGN

14.1 Requirements

- Fully responsive mobile interface
- Offline access to downloaded lessons and materials
- Auto-sync progress when connectivity is restored

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15. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full access, course management, analytics
Teacher	Create/manage courses, assessments, view analytics
Parent	View child's progress and reports
Student	Access assigned courses and content

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16. SCALABILITY

- Support thousands of concurrent learners
- Handle multiple courses, subjects, and grade levels
- Optimize performance for content delivery and analytics
- Extendable for integration with external learning platforms

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17. IMPLEMENTATION GAPS (CRITICAL FIXES)

17.1 Current Issues Identified

- Courses not published (blocking learner access)
- Limited subject and content coverage
- Insufficient assessments for evaluation
- Missing analytics dashboards
- Weak integration with other school modules

17.2 Required Fixes

- Enable course publishing workflows
- Expand subject and content library
- Increase assessment volume and automation
- Implement analytics dashboards
- Integrate with Student, Academics, and Exams modules

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18. FUTURE ENHANCEMENTS

- AI-powered personalized learning paths
- Gamification (badges, leaderboards)
- Advanced analytics and predictive insights
- Integration with external LMS platforms
- Voice-assisted learning and smart recommendations